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LOUDSPEAKER AND BASKET THEREOF

Abstract

A loudspeaker including a speaker basket. The speaker basket includes a first frame including an inner surface and an outer surface opposite the inner surface, and a second frame housed in the first frame. The second frame includes a first wall and a second wall, wherein the first wall and the second wall of the second frame engage the inner surface of the first frame.

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Background/Summary

BACKGROUND

[0001] The present invention relates to a basket for loudspeaker assemblies. A loudspeaker assembly is an electromechanical device that produces acoustic signals across a frequency range depending, at least in part, on the field strength of the permanent magnetic circuit used in the loudspeaker. It is generally accepted that the best way to increase the power handling of a

loudspeaker is to increase the surface area of the voice coil, so it can radiate/dissipate more heat. This increase in voice coil surface area almost always increases the mass of the accompanying permanent magnetic circuit. Thus, a high power transducer necessitates a heavier magnetic structure than a low power transducer. The heavier magnetic structure increases the stress on the basket of loudspeaker assemblies.

SUMMARY

[0002] The present disclosure provides, in one aspect, a loudspeaker including a speaker basket. The speaker basket includes a first frame having an inner surface and an outer surface opposite the inner surface. The speaker basket further includes a second frame housed in the first frame. The second frame includes a first wall and a second wall. The first wall and the second wall of the second frame engage the inner surface of the first frame.

[0003] The present disclosure provides, in one aspect, a speaker basket to house components of a loudspeaker. The speaker basket includes a first frame configured to be coupled to a permanent magnetic circuit at a proximal end of the first frame. The first frame includes an inner surface and an outer surface. The speaker basket further includes a second frame configured to engage the inner surface of the first frame.

[0004] The present disclosure provides, in one aspect, a loudspeaker including a permanent magnetic circuit positioned at a proximal end of the loudspeaker and a first frame of a speaker basket coupled to the permanent magnetic circuit. The first frame includes a first wall that encircles a central axis defined by the loudspeaker, a second wall that encircles the central axis and defines an angle with the first wall, and a third wall that encircles the central axis and defines an angle with the second wall. The speaker basket further includes a second frame inside of the first frame. The second frame includes a first wall that encircles the central axis, a second wall that encircles the central axis and defines an angle with the first wall, and a third wall that encircles the central axis and defines an angle with the second wall. The first wall of the first frame engages the first wall of the second frame. The second wall of the first frame engages the second wall of the second frame. The third wall of the first frame engages the third wall of the second frame.

[0005] Other features and aspects of the invention will become apparent by consideration of the following detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 is a perspective view of a loudspeaker.

[0007] FIG. 2 is a cross-sectional view of the loudspeaker through section line 2-2 on FIG. 1.

[0008] FIG. 3 is a detail view of a first frame and a second frame of a speaker basket of the loudspeaker as shown in FIG. 2.

[0009] FIG. 4 is an enlarged view of the first frame and the second frame of FIG. 3

[0010] Before any embodiments of the invention are explained in detail, it is to be understood that the invention is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The invention is capable of other embodiments and of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting.

DETAILED DESCRIPTION

[0011] FIGS. 1-2 illustrate a loudspeaker **100** according to one embodiment of the present disclosure. The loudspeaker **100** provides an auditory output from an electrical input of an external device (e.g., microphone, cellular device, sound board, etc.). The loudspeaker **100** includes a permanent magnetic circuit **112**, a spider **116**, a cone **120**, a surround **124**, a dust cover **128**, and a

voice coil **140**. The loudspeaker **100** defines a central axis A and extends from a proximal end **104** to a distal end **108**. The permanent magnetic circuit **112** is magnetically coupled to the voice coil **140** to provide linear movement, along the central axis A, as directed by the electrical input into the voice coil **140**. The dust cover **128** is positioned on the proximal end **104** of the cone **120** to protect a gap **144** and the voice coil **140** from unwanted debris. The spider **116** and the surround **124** act as a mechanical spring for the cone **120** as it moves during operation. The permanent magnetic circuit **112** and the voice coil **140** are configured to convert electrical input into a mechanical output, such that sound may be produced from the loudspeaker **100**. The permanent magnetic circuit **112** is coupled to a speaker basket **132**, such that a portion of the permanent magnetic circuit **112** is received in the basket **132**. The permanent magnetic circuit **112** includes a magnet **136**, a t-yoke **137**, a pole tip **139**, and a front plate **141**. The magnet **136** defines a generally annular shape around a portion of the voice coil **140**. The magnet **136** is magnetized along the central axis A forming proximal and distal flat surfaces of magnetic poles. The t-yoke **137**, the pole tip **139**, and the front plate **141** serve to redirect the magnetic poles in a radial direction. The magnetic poles produce a magnetic field that is concentrated across the gap **144**. The magnet **136** may be of varying size depending on the size of the permanent magnetic circuit **112** and the desired output of the loudspeaker **100**. In other embodiments, the permanent magnetic circuit **112** may have multiple magnets. The permanent magnetic circuit **112** defines the gap **144**, such that the voice coil **140** may move in/out along the central axis A. The voice coil **140** circumscribes a voice coil tube **148** that slides towards or away from the magnet **136** in the gap **144**. The movement of the voice coil **140** is dependent upon the input from the external device through a power terminal **152** located on the basket **132**. The voice coil **140** receives AC power from the external device and either propels toward the distal end **108** of the loudspeaker **100** or propels towards the proximal end **104** of the loudspeaker **100**. Movement of the cone **120**, attached to the voice coil **140**, creates pressure waves in the air. The pressure waves in the air are defined as the auditory output and can be perceived as sound by the user or audience in front of the distal end **108**.

[0012] With reference to FIGS. 1-4, the loudspeaker **100** is supported by the basket **132** to house the spider **116**, the cone **120**, and the surround **124**. The basket **132**, shown in isolation in FIG. 3, extends between a proximal end **156** and a distal end **160**. The basket **132** defines a concave shape between the proximal and distal ends **156**, **160**. The permanent magnetic circuit **112** is coupled to the proximal end **156** of the basket **132** by a plurality of fasteners **134** (FIG. 2). The distal end **160** of the basket **132** includes a plurality of case mounting apertures **164** (FIG. 1) to couple the loudspeaker **100** to a speaker enclosure (not shown). The basket **132** includes a main frame **168** and a support frame **172**. Each of the main frame **168** and the support frame **172** generally forms a stepped conical shape concentric about the central axis A of the loudspeaker **100**. As shown, the basket **132** is formed by the two nested frames: the main frame **168** and the support frame **172**. However, the basket **132** may optionally include additional frames in other constructions.

[0013] With reference to FIGS. 2-4, the main frame **168** encircles the central axis A and extends from the proximal end **156** of the basket **132** to the distal end **160** of the basket **132**. The main frame **168** defines an inner surface **176**, coupled to the spider **116** and the surround **124**, and an outer surface **180** opposite the inner surface **176**. The main frame **168** may be a stamped frame constructed from a sheet of metal (e.g., steel). Alternatively, the main frame **168** may be a die cast aluminum frame, a plastic frame, or another suitable material. The sheet forming the main frame **168** may have a thickness of at least 1.0 mm and not more than 1.2 mm in some constructions. On the proximal end **156**, the main frame **168** includes a proximal opening **184a** defining a first diameter D1 measured perpendicular to the central axis A of the loudspeaker **100**. The main frame **168** couples to the permanent magnetic circuit **112** on the proximal end **156** and a portion of the permanent magnetic circuit **112** is received within the proximal opening **184a** of the main frame **168**. The main frame **168** includes a plurality of stepped conical walls that encircle the central axis A and form an angle with an adjacent wall. More particularly, the main frame **168** includes an outer

conical wall **188a**, a connecting wall **192a**, an inner conical wall **196a**, and a proximal wall **200a**. However, the main frame **168** may optionally include additional walls in other constructions.

[0014] With reference to FIG. 4, the outer conical wall **188a** forms a conical shape around the central axis A. The outer conical wall **188a** of the main frame **168** extends from the distal end **160** of loudspeaker **100** to an upper knee **204a** with the connecting wall **192a**. The outer conical wall **188a** includes a plurality of windows **208** extending around the conical shape of the outer conical wall **188a**. The plurality of windows **208** define generally rectangular apertures in the outer conical wall **188a**. Positioned between the adjacent plurality of windows **208** is a plurality of struts **212**. The plurality of struts **212** span a portion of the outer conical wall **188a**. Specifically, the plurality of struts **212** are spaced apart from the upper knee **204a** by a distance.

[0015] The connecting wall **192a** adjoins with the outer conical wall **188a** at the upper knee **204a**. The upper knee **204a** forms a first reflex angle **206** when measured from the outer surface **180** of the main frame **168**. The connecting wall **192a** defines an annular shape around the central axis A and extends between the upper knee **204a** and an intermediate knee **216a** with the inner conical wall **196a**.

[0016] The inner conical wall **196a** forms a conical shape around the central axis A. The inner conical wall **196a** adjoins the connecting wall **192a** at the intermediate knee **216a**. The intermediate knee **216a** forms an angle α (as measured on the exterior) that is greater than 90 degrees and less than 180 degrees when measured between the outer surface **180** of the connecting wall **192a** and the inner conical wall **196a**. The inner conical wall **196a** extends between the intermediate knee **216a** and a lower knee **220a** with the proximal wall **200a**.

[0017] The proximal wall **200a** defines an annular shape and is generally parallel with the connecting wall **192a**. The proximal wall **200a** adjoins the inner conical wall **196a** at the lower knee **220a**. The lower knee **220a** defines a second reflex angle **210** when measured between the inner conical wall **196a** and the proximal wall **200a** on the outer surface **180** of the main frame **168**. The proximal wall **200a** includes a first plurality of mounting apertures **224** centered around the central axis A. Each mounting aperture **224** receives one of the plurality of fasteners **134** to couple the main frame **168** to the permanent magnetic circuit **112**. The permanent magnetic circuit **112** includes a plurality of fastener bores **138** (FIG. 2) to receive the plurality of fasteners **134**.

[0018] With reference to FIGS. 2-4, the support frame **172** of the basket **132** is positioned inside of the main frame **168** and may be a stamped frame constructed from a sheet of metal (e.g., steel). Alternatively, the support frame **172** may be a die cast aluminum frame, a plastic frame, or another suitable material. The sheet forming the support frame **172** may have a thickness of at least 1.0 mm and not more than 1.2 mm. In some constructions, the sheet thickness of the support frame **172** is equal to the sheet thickness of the main frame **168**. The support frame **172** defines an interior **228** and an exterior **232** opposite the interior **228**. The exterior **232** of the support frame **172** engages the inner surface **176** of the main frame **168**. The exterior **232** of the support frame **172** may be bonded via an adhesive to the inner surface **176** of the main frame **168**. The support frame **172** encircles the central axis A and extends between a proximal end **236** and a distal end **240**. The proximal end **236** includes a proximal opening **184b** to receive a portion of the permanent magnetic circuit **112** and defines a second diameter D2 measured perpendicular to the central axis A of the loudspeaker **100**. The first diameter D1 of the main frame **168** is equal to the second diameter D2 of the support frame **172**. The distal end **240** of the support frame **172** is positioned in the distance between the plurality of struts **212** and the upper knee **204a** of the main frame **168**. The support frame **172** generally mirrors the plurality of walls of the main frame **168** and therefore includes a corresponding outer conical wall **188b**, a corresponding connecting wall **192b**, a corresponding inner conical wall **196b**, and a corresponding proximal wall **200b**. The support frame **172** further includes a corresponding upper knee **204b**, a corresponding intermediate knee **216b**, and a corresponding lower knee **220b**.

[0019] With reference to FIG. 4, the outer conical wall **188b** forms a conical shape around the

central axis A and is parallel and in engagement with the inner surface **176** of the outer conical wall **188a** of the main frame **168**. The outer conical wall **188b** defines the distal end **240** of the support frame **172** and therefore does not extend past the plurality of struts **212** of the main frame **168**. The outer conical wall **188b** adjoins the connecting wall **192b** at the upper knee **204b**. The upper knee **204b** defines the first reflex angle **206** when measured from the exterior **232** of the outer conical wall **188b** and the connecting wall **192b** of the support frame **172**.

[0020] The connecting wall **192b** of the support frame **172** defines an annular shape and is parallel and in engagement with the inner surface **176** of the connecting wall **192a** of the main frame **168**. The connecting wall **192b** engages with the spider **116** on the interior **228** and the connecting wall **192a** of the main frame **168** on the exterior **232**. The connecting wall **192b** extends from the upper knee **204b** to the intermediate knee **216b** formed with the inner conical wall **196b**. The intermediate knee **216b** defines the angle α when measured from the exterior **232** of the connecting wall **192b** and the inner conical wall **196b** of the support frame **172**.

[0021] The inner conical wall **196b** of the support frame **172** forms a conical shape around the central axis A and is parallel and in engagement with the inner surface **176** of the inner conical wall **196a** of the main frame **168**. The inner conical wall **196b** extends from the intermediate knee **216b** to the lower knee **220b** formed with the proximal wall **200b**. The lower knee **220b** defines the second reflex angle **210** when measured from the exterior **232** of the inner conical wall **196b** and the proximal wall **200b** of the support frame **172**.

[0022] The proximal wall **200b** of the support frame **172** defines an annular shape and is parallel with the connecting wall **192b** of the support frame **172**, the connecting wall **192a** of the main frame **168**, and the proximal wall **200a** of the main frame **168**. The proximal wall **200b** is in engagement with the inner surface **176** of the proximal wall **200a** of the main frame **168**. The proximal wall **200b** includes a second plurality of mounting apertures **244** centered around the central axis A. Each of the second plurality of mounting apertures **244** is concentric with one of the first plurality of mounting apertures **224**. The plurality of fasteners **134** extend through the first plurality of mounting apertures **224** and the second plurality of mounting apertures **244** to couple the permanent magnetic circuit **112** to the basket **132**.

[0023] In an alternative embodiment, the basket **132** may include an additional frame positioned inside of the main frame **168** or the support frame **172**. The additional frame may be a stamped steel frame, a die cast aluminum frame, a plastic frame, or another suitable material. The addition of another frame may further enhance the strength of the basket **132** and may enable use with even larger transducers.

[0024] A transducer used in a loudspeaker includes a voice coil of varying diameter based on the desired maximum auditory output. As the voice coil increases in diameter, the size and weight of the magnet and steel parts used in the permanent magnetic circuit also increases. Typically, a low power transducer uses a stamped steel frame with a voice coil diameter less than 76.2 mm (3 inches). Typically, a high power transducer uses a die cast aluminum frame with a voice coil diameter greater than 76.2 mm (3 inches). Stamped steel frames are advantageous over die cast aluminum frames, as stamped steel frames weigh less and are more cost-effective than die cast aluminum frames. However, a simple stamped steel frame according to the prior art may not support the weight of a high power transducers permanent magnetic circuit. As seen in FIGS. 3-4, the intermediate knee **216a, 216b** forms the angle α . If a stamped steel frame of a single sheet according to the prior art is coupled to a high power transducer on its proximal end, the stamped steel frame may distort (e.g., causing the angle α to become asymmetrical about the conical shape), thereby damaging the transducer. The general use of the transducer by the consumer or the shipping process from the manufacture to the consumer may cause the distortion of the stamped steel frame due to the added weight. If such bending of the stamped steel frame occurs, the gap **144** of the permanent magnetic circuit **112**, in which the voice coil **140** moves, may decrease. If the clearance for the voice coil **140** defined by the gap **144** decreases from its original condition, the voice coil

140 may engage a portion of the permanent magnetic circuit **112**, which may damage the loudspeaker **100**. Therefore, the addition of the support frame **172** near the proximal end **156** of the main frame **168** may provide suitable longevity with the use of a high power permanent magnetic circuit, without incurring excess weight and cost associated with a die cast aluminum frame. [0025] Various features of the invention are set forth in the following claims.

Claims

1. A loudspeaker comprising: a first frame including an inner surface and an outer surface opposite the inner surface; and a second frame housed in the first frame, the second frame including a first wall and a second wall, the first and second frames forming a speaker basket, wherein the first wall and the second wall of the second frame engage the inner surface of the first frame.
2. The loudspeaker of claim 1, wherein the first frame is a first stamped steel frame, and the second frame is a second stamped steel frame.
3. The loudspeaker of claim 1, wherein the first wall and the second wall of the second frame define an angle that matches an angle defined by the inner surface of the first frame.
4. The loudspeaker of claim 3, wherein the angle is greater than 90 degrees and less than 180 degrees when measured between an exterior of the first wall and the second wall.
5. The loudspeaker of claim 1, further comprising a permanent magnetic circuit, wherein the first frame includes a first plurality of mounting apertures, and the first frame is coupled to the permanent magnetic circuit via a plurality of fasteners that extend through the first plurality of mounting apertures.
6. The loudspeaker of claim 5, wherein the second frame includes a second plurality of mounting apertures, and the plurality of fasteners extend through the second plurality of mounting apertures.
7. The loudspeaker of claim 1, wherein each of the first and second frames has a stepped conical shape concentric about a central axis of the loudspeaker.
8. The loudspeaker of claim 5, wherein the first frame includes a first opening on a proximal end of the loudspeaker and the second frame includes a second opening on the proximal end, and wherein the first opening and the second opening receive a portion of the permanent magnetic circuit.
9. The loudspeaker of claim 8, wherein the first opening defines a first diameter measured perpendicular to a central axis of the loudspeaker, and the second opening defines a second diameter measured perpendicular to the central axis of the loudspeaker, wherein the second diameter is equal to the first diameter.
10. The loudspeaker of claim 1, wherein the second frame is constructed from a sheet and has a thickness of at least 1.0 mm and not more than 1.2 mm.
11. A speaker basket to house components of a loudspeaker, the speaker basket comprising: a first frame configured to be coupled to a permanent magnetic circuit at a proximal end of the first frame, the first frame including an inner surface and an outer surface; and a second frame configured to engage the inner surface of the first frame.
12. The speaker basket of claim 11, wherein the second frame is bonded to the first frame via an adhesive.
13. The speaker basket of claim 11, wherein the first frame includes a plurality of struts between the proximal end and a distal end of the first frame, and wherein the second frame has a distal end between the plurality of struts and the proximal end of the first frame.
14. The speaker basket of claim 11, wherein the first frame and the second frame are constructed from a sheet to form a stepped cone geometry.
15. A loudspeaker comprising: a permanent magnetic circuit positioned at a proximal end of the loudspeaker; a first frame coupled to the permanent magnetic circuit, the first frame including: a first wall that encircles a central axis defined by the loudspeaker; a second wall that encircles the central axis and defines an angle with the first wall, and a third wall that encircles the central axis

and defines an angle with the second wall; and a second frame inside of the first frame and forming a speaker basket therewith, the second frame including: a first wall that encircles the central axis, a second wall that encircles the central axis and defines an angle with the first wall, and a third wall that encircles the central axis and defines an angle with the second wall, wherein the first wall of the first frame engages the first wall of the second frame, wherein the second wall of the first frame engages the second wall of the second frame, and wherein the third wall of the first frame engages the third wall of the second frame.

16. The loudspeaker of claim 15, wherein the first frame includes a fourth wall that encircles the central axis and defines an angle with the third wall.

17. The loudspeaker of claim 16, wherein the second frame includes a fourth wall that encircles the central axis and defines an angle with the third wall.

18. The loudspeaker of claim 17, wherein the fourth wall of the first frame engages the fourth wall of the second frame.

19. The loudspeaker of claim 15, wherein a fourth wall of the first frame includes a first plurality of mounting apertures, and the first frame is coupled to the permanent magnetic circuit via a plurality of fasteners that extend through the first plurality of mounting apertures.

20. The loudspeaker of claim 19, wherein a fourth wall of the second frame includes a second plurality of mounting apertures, and the second frame is coupled to the permanent magnetic circuit via the plurality of fasteners that extend through the second plurality of mounting apertures.
